

Portfolio Optimization Class Project Decomposition

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Abstract

This is a decomposition for one of the most interesting course projects I took in my undergraduate (September 2025). This project focuses on applied modern portfolio theory and rigorous Excel usage to construct a 30 stock portfolio, using the Dow Jones Industrial price-weighted index as a reference point, where, we must build the best possible portfolio with the optimal risk-adjusted returns. In this decomposition paper, the mathematics and theoretical concepts are broken down and applied to each step of the project followed by the final results and discussion. Ultimately, this paper is aimed for employers, professionals, and any other finance practitioners who happen to be interested in reading. The Excel file can be found and downloaded on the project page for proof of work. Most importantly, I hope it is an insightful and interesting read.

1 Introduction

The entire premise of the project was to restructure the Dow Jones Industrial Index ($\hat{DJI}$) into a optimal risky portfolio; completely reweighting the index from its original a price-weighted formation into a portfolio that maximizes its risk-to-reward ratio, Sharpe Ratio, and delivers the most optimal weighting for all 30 equities. The final reweighting includes long and short positions for each individual stock that generates the highest possible risk-adjusted return. We will proceed chronologically in each phase of the construction of the optimal portfolio. Following, we will discuss the data extraction and aggregation methods.

2 Price Data Extraction

To begin, the entire asset universe is of course limited to public-traded equities and bound by the 30 stock set from the Dow Jones Industrial Index. The most important component of the project is obviously the price data that will be fed into it. The higher quantity of data, the more accurate the calculations become. The primary source for data was extracted from Wharton Research Data Services with table 1 displaying the list of equities. Additionally, Capital IQ Pro was utilized for complementary research but not pertinent to this paper. Students were given the option to choose daily, weekly, or even monthly returns as well as the amount of years. I settled to download daily price data for all 30 equities from January 1st, 2000 until December 31st, 2024, this amounted to over 190,000 entries, downloaded and dumped into Excel; the raw data had three columns: date, ticker symbol, and the return (including six decimal places). The next step was to aggregate all the data into an incredibly large pivot table in a way where there were now 31 columns, which included the 30 ticker symbols and corresponding date. Essentially, the pivot table just transformed our initial raw data sheet into a more organized table. Also, at the last of cell of each stock, the total daily return was computed for the each stock using the summation function. At this point, we can begin constructing matrices and start performing relevant computations.

Table 1: **Dow Jones Industrial List**

Company Name	Ticker Symbol
Apple Inc.	AAPL
Amgen Inc.	AMGN
Amazon.com Inc.	AMZN
American Express Company	AXP
The Boeing Company	BA
Caterpillar Inc.	CAT
Salesforce, Inc.	CRM
Cisco Systems, Inc.	CSCO
Chevron Corporation	CVX
The Walt Disney Company	DIS
The Goldman Sachs Group, Inc.	GS
The Home Depot, Inc.	HD
Honeywell International Inc.	HON
International Business Machines Corporation	IBM
Johnson & Johnson	JNJ
JPMorgan Chase & Co.	JPM
The Coca-Cola Company	KO
McDonald's Corporation	MCD
3M Company	MMM
Merck & Co., Inc.	MRK
Microsoft Corporation	MSFT
NIKE, Inc.	NKE
NVIDIA Corporation	NVDA
The Procter & Gamble Company	PG
The Sherwin-Williams Company	SHW
The Travelers Companies, Inc.	TRV
UnitedHealth Group Incorporated	UNH
Visa Inc.	V
Verizon Communications Inc.	VZ
Walmart Inc.	WMT

2.1 Standard Return Matrix

The next step, was to copy the pivot table and paste onto a new sheet and begin editing it to construct our standard return matrix. Table 2 illustrates a preview of the standard return matrix that is 4228x31 (rows by columns) and is a preliminary but crucial table in this project. It captures all the daily returns across all 30 equities with the respective dates, with the last two rows of this matrix have a total return and average return of each equity. The total the same summation from the earlier pivot table (sum of all the daily returns for a particular stock), while the average is a simple arithmetic return of all the daily returns for each stock, which is also equal to the investor's expected return. Although, one major adjustment was made to this matrix, that is the reduction of the sample size; the sample now begins on March 20th, 2008 and all the daily return data since January 1st, 2000 was purged. This was done because of numerous blank cells in the return data for certain equities. The loss of eight years worth of data will reduce our sample but worth the hassle as blank cells would result in incalculable operations later on. The arithmetic mean can be calculated simply as stated, where, n is the total number of daily returns for each equity and R_n is each daily return:

$$E(R_i) = \frac{(\sum_{t=i}^n R)}{n} = \frac{R_1 + R_2 + R_3 + \dots + R_{daily}}{n}$$

Note for table 2, that R_{daily} is the daily return of respective stock, T_i representing each date until the end of the sample, and S_i being defined as the other 28 stocks, with the real matrix of course extending much further and longer.

Table 2: Standard Return Matrix

Dates	AAPL	AMGN	$S_{i=1,2,3,\dots,30}$
3/20/2008	0.027763	0.003503	R_i
3/24/2008	0.046972	0.02169	R_{i+1}
T_i	...	...	...
Total (ΣR_i)	0.00123	0.000475	...
Average ($E(R_i)$)	0.00117	0.000662	...

2.2 Abnormal Return Matrix

The next step is to construct a unexpected or abnormal return matrix using the data from the previous standard return matrix. The abnormal return matrix is constructed by scratch with each cell now containing the abnormal return for each stock, instead of daily returns. Table 3 provides the preview, with the actual matrix carrying dimensions of 4226x31. The abnormal return which computed as stated below as such:

$$AR = R_{daily} - E(R_i)$$

Abnormal return captures the real profit or loss that the investor gains for each security depending if a positive or negative is returned. The reason that this abnormal matrix was constructed is because it is a prerequisite step to build the sum of squared deviations matrix for a vector of equity returns, which is required to build our eventual covariance matrix.

Table 3: Unexpected (Abnormal) Return Matrix, $N_{Columns} = 4225$

Dates	AAPL	AMGN	$S_{i=1,2,3,\dots,30}$
3/20/2008	0.02659	0.00284	...
3/24/2008	0.04580	0.02103	...
3/25/2008	0.00922	-0.00408	...
...	...	...	...
T_i	AR_i	AR_{i+1}	...

2.3 Sum of Squared Deviations Matrix

Using the abnormal return matrix, we begin to build our next matrix. To construct this matrix, the sum of squared deviations (SSD) needs to be converted into in matrix notation. It can be decomposed as follows:

Let $\mathbf{R}_i$ be a $T \times 1$ vector of observed returns for security i over T periods, and let $\mathbb{E}[R_i]$ be its expected (mean) return. The deviation vector $\mathbf{u}_i$ is defined as:

$$\mathbf{AR}_i = \mathbf{R}_i - \mathbb{E}[R_i]\mathbf{1} = \begin{bmatrix} R_{i1} - \mathbb{E}[R_i] \\ R_{i2} - \mathbb{E}[R_i] \\ \vdots \\ R_{iT} - \mathbb{E}[R_i] \end{bmatrix}$$

The sum of squared deviations is calculated by taking the inner product of this deviation vector with itself (in Excel, this means multiplying the transpose of the vector by the vector itself). The mathematical layout is as stated, where, AR'_i is the transposed vector of itself.

$$\mathbf{AR}'_i \mathbf{AR}_i = \{\mathbf{R}_i - \mathbb{E}[R_i]\}' \{\mathbf{R}_i - \mathbb{E}[R_i]\}$$

Expanded into matrix components, the multiplication is expressed as:

$$\sum_{t=1}^T (R_{it} - \mathbb{E}[R_i])^2 = \begin{bmatrix} R_{i1} - \mathbb{E}[R_i] & R_{i2} - \mathbb{E}[R_i] & \dots & R_{iT} - \mathbb{E}[R_i] \end{bmatrix} \begin{bmatrix} R_{i1} - \mathbb{E}[R_i] \\ R_{i2} - \mathbb{E}[R_i] \\ \vdots \\ R_{iT} - \mathbb{E}[R_i] \end{bmatrix}$$

The sum of squared deviations formula can be simplified to:

$$SSD = \sum_{t=1}^T (R_{i,t} - E(R_i))^2$$

The transpose matrix operation can easily be done in Excel, by simply converting the column vector into a row of vector, so each cell can be multiped against itself which is essentially acts as a squaring operation. Table 3 is a preview to display what the sum of squared deviation matrix looks like within the project file.

Table 4: Sum of Squared Deviations Matrix

APPL	AMGN	AMZN	S _i
1.55983	0.48445	0.94606	...
0.48445	1.14676	0.53033	...
0.94606	0.53033	2.25581	...
...	...	...	...
SSD _{APPL}	SSD _{AMGN}	SSD _{AMZN}	SSD _{S_i}

With the SSD calculated, we are able to compute the variance, volatility (standard deviation), and mostly importantly, the covariance matrix.

2.4 Covariance Matrix

The Covariance matrix is the skeleton of this portfolio management project, as it is foundational to modern portfolio theory. Covariance allows us to measure the relationship in how each security move together, this discovery is crucial to compute the optimal weights of the portfolio. Building the fourth and final matrix, we once again use the previous matrix as a derivative for our calculations. The variance of every single corresponding security must be computed by dividing every cell from the SSD matrix by the degrees of freedom in the sample, $N_{Columns} - 1 = 4224$, giving us the variances for every possible combination for every two securities. The variance formula is stated as such:

$$\frac{SSD_{i=S_1, S_2, S_3, \dots, S_{30}}}{N_{Columns} - 1} = \sigma_i^2$$

To build the sample covariance matrix, denoted by Σ , we compute the outer product of this matrix with its transpose and scale it by the degrees of freedom, in the end, resulting in a 30x30 covariance matrix with a total of 900 cells in the Excel file.

$$\Sigma = \begin{bmatrix} \sigma_{1,1}^2 & \sigma_{1,2} & \dots & \sigma_{1,30} \\ \sigma_{2,1} & \sigma_{2,2}^2 & \dots & \sigma_{2,30} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{30,1} & \sigma_{30,2} & \dots & \sigma_{30,30}^2 \end{bmatrix}$$

Next, the formula for total **portfolio variance** (σ_p^2) using a vector of security weights ($\mathbf{w}$) is given by (note, that the weights of portfolio will be calculated using Excel's Data Solver function, more on this later):

$$\sigma_p^2 = \mathbf{w}' \Sigma \mathbf{w}$$

2.5 The Risk-free Rate

Before, calculating total return, standard deviation, and Sharpe ratio of the portfolio, we must establish a key market parameter, the risk-free rate. This is the required rate of return that the portfolio must surpass in order for any rational investor to take on. I chose a relatively high risk-free rate of $R_f = 4\%$. This is converted in a daily rate to match the portfolio, the following calculation is stated, where, only trading days will taken into account.

$$\text{Daily}(R_f) = (1 + R_f)^{1/252} - 1$$

Table 5 shows the data for the risk-free rate which is a benchmark for the portfolio's performance.

Table 5: Risk-Free Metrics

Risk-free	(%)
R_f	4.00%
Daily(R_f)	0.0156%
σ_f	Risk-free

2.6 Portfolio Weighting, Optimization and Results

This is the final stage of the project, where the final computations will be made, broken down into three main portfolios. The equal weighting portfolio (mainly as a reference), global minimum variance portfolio which offers the lowest possible risk and volatility, and the optimal risky portfolio which offers the maximum Sharpe ratio. Figure 1 displays the capital market line (CML); this graph demonstrates the best possible return given the certain magnitude of risk. Clearly, the optimal risky portfolio offers the best risk-adjusted returns possible.

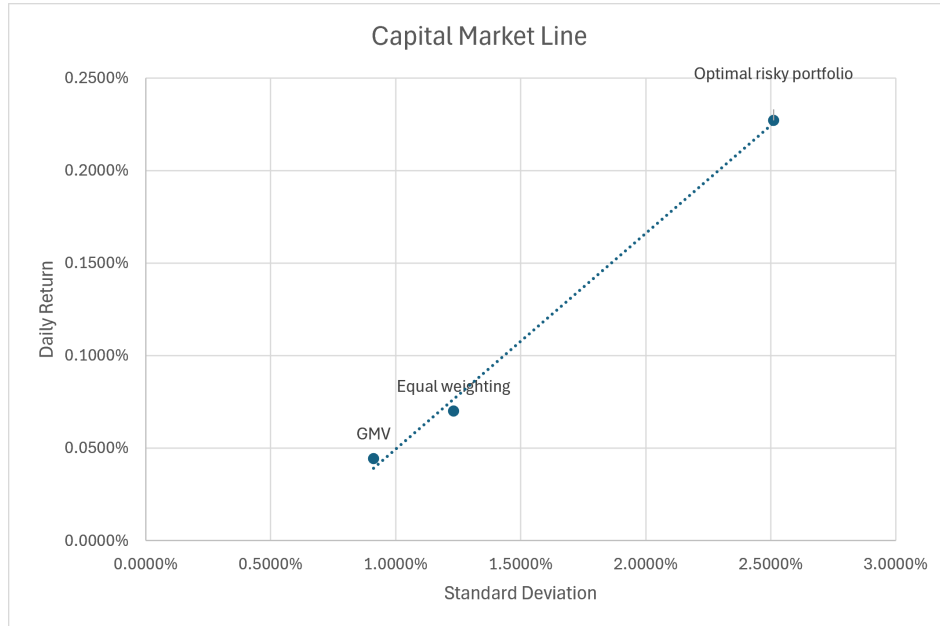


Figure 1: CML With Three Final Portfolios

In Excel, there a tremendously helpful optimization tool called the Data Solver function that finds the maximum or minimum value given inputted variables with certain constraints outlined by the user. In this context, it was used to calculate the weights for global minimum variance and optimal risky portfolio. First, to solve for the GMV portfolio, we set the objective to find the minimum or lowest possible standard deviation, setting the variable cells (solving for weights) to change, and subjecting the constraint as the total sum of the weights being equal to one. The same process is done to solve for

the optimal risky portfolio but the only difference is changing the first parameter, where, the objective is set to find the maximum return, holding everything else the same. After computing, Excel will spit out the correct weights for each security. The equal weighting portfolio can be calculated manually, by obviously just dividing the portfolio evenly across (1/30), resulting in each weighting being 3.33% for each stock. All three portfolios are then inputted into a formal table, refer to the table 8 in the appendix to observe all of the weights for all three portfolios. Conclusively, we can clearly see that the optimal risky portfolio experiences the highest daily return and standard deviation, but, falls in between in terms of the steepness of its slope.

Table 6: Table Note for Figure 1

	Global Minimum Variance (GMV)	Equal Weighting	Optimal Risky Portfolio
Daily(R_p)	0.0444%	0.0701%	0.2271%
σ_p	0.9113%	1.2314%	2.5111%
Slope	3.1638%	8.4245%	4.4288%

Converting the daily returns and standard deviation into their annualized form gives us the final output for this project. The most ideal portfolio provides an annual return of 77.12% with a commensurate volatility of 39.86%, now, on its surface, it may seem like a truly exceptional portfolio but there are of course underlying assumptions to be made. On the annual return frontier, we assume independence of daily returns (formally known as random walk) which means that the return performance of today does not affect tomorrow and vice versa, this presumes that daily returns are independent. In the real markets, this is clearly not the case and price momentum is a real phenomenon observed due to human behavior and macroeconomic conditions being prime drivers in the equity markets, sometimes causing irrational price movements. Secondly, is the constant-expected return framework, which states that all securities within the portfolio have a stable drift throughout the year without the introduction of systematic risk, for example, disrupted supply chains affecting trade. In order to simulate such events (external risk), we would need to implement a stress-testing method like a Monte Carlo simulation, for now, this factor is omitted. The last assumption is that there are no transaction costs and that all proceeds of income (from dividends) are assumed to be reinvested back into the portfolio. Conversely, the assumptions for standard deviation are quite similar, where, the volatility of each security is assumed to be consistent for the entire year and variance also scales under the random walk assumption. Finally, the optimal risky portfolio has short positions that may be unideal in a real investment setting due to the inherent riskiness of such leveraged positions, for instance, there is a -19.93% short position on NIKE, Inc. (NKE) which would be outside of most portfolio manager's risk tolerances.

Table 7: Annual Returns and Standard Deviations

Annual	Global Minimum Variance (GMV)	Equal Weighting	Optimal Risky Portfolio
R_p	31.16%	19.31%	77.12%
σ_p	22.13%	19.55%	39.86%

3 Discussion and Theory

The main takeaway from the project was applying modern portfolio theory in an applicable way, constructing a portfolio using real historical price data, mean-variance analysis, graphing the capital market line, and assembling the covariance matrix as a fundamental building block to this project, as it was necessary to discover how each equity correlations with each other. Offsetting unnecessary risk without sacrificing returns is a key pillar in portfolio construction and the covariance matrix made this possible. Within the 30 equity list, certain equities are bound to move or less in relation to one another, for instance, VISA Inc. and Apple Inc. would both decline if consumer spending fell dramatically but Apple might suffer more if semiconductor chips exploded in price. Hence, the covariance of VISA and Apple is quite small at 0.0173%, meaning they have a weak directional linear relationship. Furthermore, the fact that this project only contains one asset class with a multitude of different equities means

that inevitably we can not hedge effectively against market risk (systematic risk). No matter how uncorrelated various of the stocks may be to one another which would provide great idiosyncratic risk prevention, greater macroeconomic conditions would prevail and possibly causing a decline for all securities, wiping out the performance if a bad enough exogenous shock were to occur. For instance, if the Federal Reserve were to enact an emergency rate cut of 100 basis points, this would shock the entire equity market, causing our entire portfolio to hemorrhage. To hedge, we would obviously need to introduce new assets, perhaps fixed-income securities or swap futures. Illustrated in figure 2, as the number of assets (or in our case, number of equities increases), total risk can be reduced but to a certain extend. The limit is reached when the horizontal asymptote is reached, representing systematic risk caused by shocks can shake the entire economy, subsequently hurting all underlying markets.

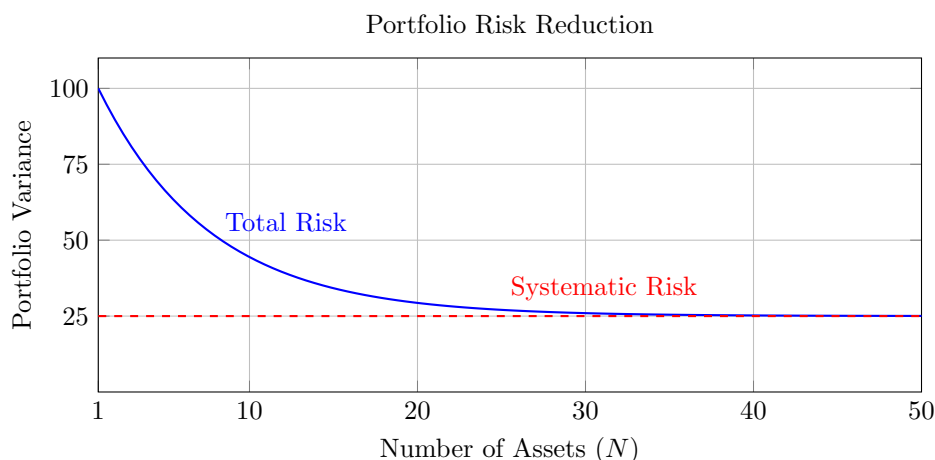


Figure 2: Simplified curve tracking portfolio risk reduction toward its systematic covariance limit.

3.1 Reflection and Notes for Future Projects

As methods of quantitative finance become more incorporated in traditional portfolio and fundamental investing, the means to test investment strategies becomes more robust. For future portfolio construction projects, proceeding to a stress-testing phase where parameters can be adjusted would be invaluable for scenario analysis. As aforementioned, a Monte Carlo simulation would allow us to test and adjust systematic risk, simulate bear or bull markets, and correlate volatility from previous to consecutive days to eliminate the random walk assumption. Bootstrapping techniques can also be utilized to test resilience of the portfolio in millions of different possible outcomes through resampling. Furthermore and beyond testing, the process in building of the portfolio can even be modified, utilizing machine learning to create multiple variations of the optimal portfolio by adding and or removing new or existing equities, overall, mitigating risk whilst retaining or increasing risk-adjusted returns. Moreover, besides different implementation methods, building a multi-asset portfolio with a more diverse set of instruments like options, swaps, and currencies would be greatly more interesting to construct in the future. I can say confidently that this was a solid project that introduce into the fascinating world of portfolio management. On a final note, this project was done fully mechanistically, all computed with Excel fed by raw data, utilizing pivot tables, various functions (MMULT, TRANSPOSE, etc.), and overall contributed as valuable experience that enhanced my proficiency with the software.

4 Conclusion

In the end, I hope this decomposition was enjoyable to read and somewhat easy to follow. The goal was meant to capture the process of applying finance theory to a real applicable project, in this case, in the context of portfolio management. I hope to endeavor on more portfolio projects in the future that only scale up in complexity. In the concurrent finance industry, quantitative methods will continue to grow and permeate, therefore, it only makes sense to take on these more mathematical approaches and understand them deeply.

For any inquires, concerns, or feedback, please do not hesitate to reach out to me, thanks again for reading!

5 Appendix

Table 8: Portfolio Weights

Ticker	GMV	Optimal Risky Portfolio	Equal Weights
AAPL	0.0000	0.4787	0.0333
AMGN	0.0000	0.1246	0.0333
AMZN	0.0000	0.2592	0.0333
AXP	0.0000	−0.0206	0.0333
BA	0.0000	−0.1189	0.0333
CAT	0.0000	0.2252	0.0333
CRM	0.0000	−0.0032	0.0333
CSCO	0.0000	−0.5190	0.0333
CVX	0.0000	−0.1441	0.0333
DIS	0.0000	−0.2551	0.0333
GS	0.0000	−0.3116	0.0333
HD	0.0000	0.2865	0.0333
HON	0.0000	−0.0858	0.0333
IBM	0.0360	−0.2666	0.0333
JNJ	0.2487	−0.0782	0.0333
JPM	0.0000	0.1900	0.0333
KO	0.1348	−0.0914	0.0333
MCD	0.1710	0.3160	0.0333
MMM	0.0000	−0.3866	0.0333
MRK	0.0000	−0.0086	0.0333
MSFT	0.0000	−0.0337	0.0333
NKE	0.0000	−0.1993	0.0333
NVDA	0.0000	0.3748	0.0333
PG	0.1080	−0.0514	0.0333
SHW	0.0012	0.4165	0.0333
TRV	0.0000	0.1260	0.0333
UNH	0.0000	0.2878	0.0333
V	0.0000	0.3083	0.0333
VZ	0.1150	−0.1491	0.0333
WMT	0.1854	0.3296	0.0333
Total	1.0000	1.0000	1.0000

Note: Negative values indicate a short position.